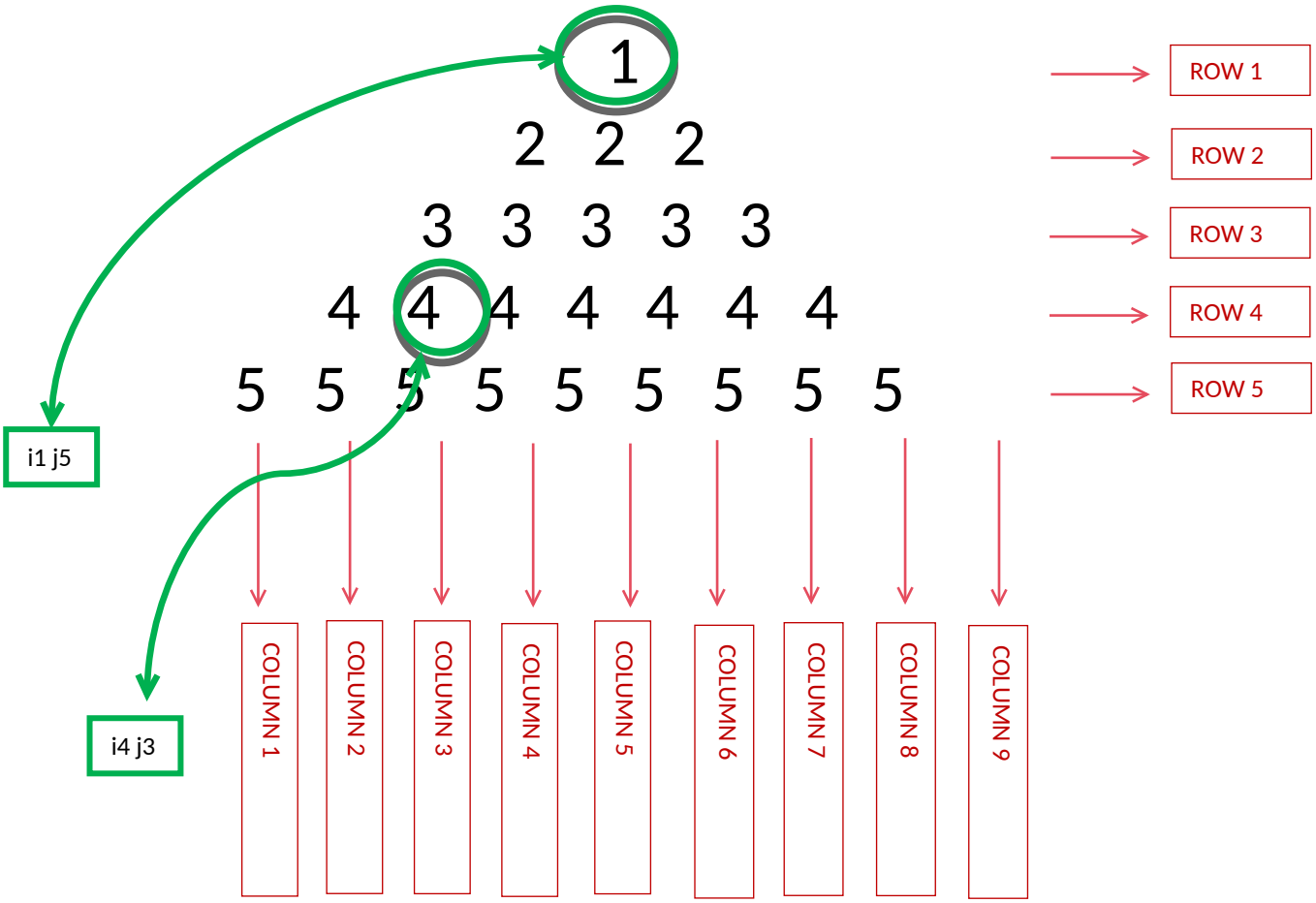
WEEK 01 MODULE
Programming Practice

TOPIC : 2 ‘ Number Pattern’

10. SAME NUMBER PYRAMID



1. Problem Analysis :-

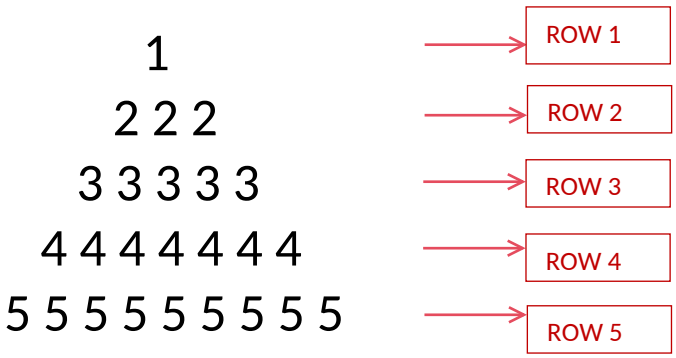
i. Pattern Structure:

- The output forms a centered same number pyramid.

- Unlike the Triangle and Inverted Triangle patterns, the number are not aligned to the left side this is aligned around a central vertical axis.

- spaces are printed before the number so that the pattern appears centered.

- For n=5



ii. Execution Flow:

Step 1: Observe the Grid Concept:

- For n = 5

1

2 2 2

3 3 3 3 3

4 4 4 4 4 4 4

5 5 5 5 5 5 5 5 5

The pattern grows symmetrically from the center.

As we move downward:

- The number of spaces decreases.
- The number increases.

Both of these changes occur simultaneously.

Therefore, this pattern requires two different calculations in every row.

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

-The pattern can be viewed as two regions:

✦ **Region 1: Leading Spaces**

These spaces shift the stars toward the center.

✦ **Region 2: Stars**

These form the visible pyramid.

For n = 5

Row 1 : [ ][1]

Row 2 : [ ][2 2 2]

Row 3 : [ ][3 3 3 3 3]

Row 4 : [ ][4 4 4 4 4 4 4]

Row 5 : [][5 5 5 5 5 5 5 5 5]

As the row index increases:

- Spaces decrease by 1.
- number stays constant in each row.

-> Row Analysis :-

Row Index (I)	1	2	3	4	5
Spaces	4	3	2	1	0
number	1	3	5	7	9
Printed value	1	222	33333	4444444	555555555

The pyramid is controlled by two formulas:

Spaces = n - i - 1

Number = 2 × i + 1

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row, calculate the number of leading spaces required.

- The number of spaces depends on the current row index.
- The relationship is:

- Spaces = n - i - 1

Step 4: Traverse the required number of positions and print number.

- This shifts the number toward the center.
- The number of spaces decreases as the row index increases.

Step 5: After printing spaces, calculate the number required.

- The relationship is:

- Number = 2 × i + 1

Step 6: Traverse the required number of positions and print numbers.

- The qyt. number increases by 2 in every row but number remains same throughout the row.

Step 7: After all number of the current row have been printed, move the cursor to the next line using System.out.println();

- This ensures that the next row starts on a new line.

Step 8: Repeat the process for all rows until the outer loop terminates.

2. Condition Analysis

No conditional statements are required.

There is no need for if(...)

- Because every position visited by the space loop prints a space.

- Every position visited by the number loop prints a number.

- The pattern is controlled entirely through loop limits.

3. Core Logic Transformation

Pyramid Pattern:-

for(int j = 0; j < 2 * i + 1; j++)

- Meaning:

quantity of numbers increase by 2 in every row.

Additionally :-
for(int j = 0; j < n - i - 1; j++)

- controls the alignment of the pattern.

4. Program :-

```
package week1_module;
import java.util.Scanner;
public class NumberPyramid {

	public static void main(String[] args) {
	Scanner sc = new Scanner(System.in);

	System.out.print("Enter the number of rows: ");
	int n = sc.nextInt();

		for(int i = 0; i < n; i++) {

			// Print spaces
			for(int j = 0; j < n - i - 1; j++) {
				System.out.print(" ");
			}

			// Print number
			for(int j = 1; j <= 2 * i + 1; j++) {
				System.out.print( i+1+ " ");
			}

			System.out.println();
		}

		sc.close();
	}
}
```

For pyramids, using **0-based indexing (**i = 0**)** ie formulas become:

Spaces = n - i - 1
Numbers = 2 * i + 1

If you force 1-based indexing, the formulas become :

Spaces = n - i
Numbers = 2 * i - 1

5. Program Output :-

Enter the number of rows: 5

```
            1
        2 2 2
      3 3 3 3 3
    4 4 4 4 4 4 4
5 5 5 5 5 5 5 5 5
```

6. **ScreenShot :-**

i. Program :-

NumberPyramid.java | SameNumberPyramid.java

```
1  package week1_module;
2  import java.util.Scanner;
3  public class SameNumberPyramid {
4
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print("Enter the number of rows: ");
9          int n = sc.nextInt();
10
11         for(int i = 0; i < n; i++) {
12
13             // Print spaces
14             for(int j = 0; j < n - i - 1; j++) {
15                 System.out.print(" ");
16             }
17
18             // Print numbers
19             for(int j = 1; j <= ( 2 * i + 1) ; j++) {
20                 System.out.print(i+1 + " ");
21             }
22
23             System.out.println();
24         }
25
26         sc.close();
27     }
28 }
29
```

ii. Program's Output :-

Problems | Javadoc | Declaration | Console

```
<terminated> SameNumberPyramid [Java Application] /Library/Java/JavaVirtual
Enter the number of rows: 5
    1
   2 2 2
  3 3 3 3 3
 4 4 4 4 4 4 4
5 5 5 5 5 5 5 5 5
```